

M.S. NAZ HIGH SCHOOL

Excellence in Matriculation & Academic Leadership | Est. 1989 | BISE Gujranwala Code: 112199

Single National Curriculum (SNC) & PECTAA Standards Aligned - Session 2026-2027

CLASS 9 (9TH MATRIC) - BIOLOGY

Official Syllabus, Board Paper Scheme & High-Yield Examination Revision Notes

Group: Science Group | Total Marks: 75 Marks (Theory: 60 | Practical: 15) | Time: 2 Hours 45 Minutes

SECTION 1: BISE GUJRWALA EXAMINATION BLUEPRINT & WEIGHTAGE

- **Section A: Objective (MCQs):**
12 Compulsory MCQs (1 Mark each = 12 Marks).
- **Section B: Short Questions:**
Attempt 15 out of 24 Short Questions (3 sub-sections: Q2, Q3, Q4: 5/8 each = 30 Marks).
- **Section C: Long Questions:**
Attempt 2 out of 3 Long Questions (Q5, Q6, Q7: 9 Marks each = 18 Marks).
- **SLO Cognitive Weightage:**
Knowledge: 50% | Understanding: 35% | Biological Diagram Labeling & Application: 15%.

SECTION 2: CHAPTER-WISE HIGH-YIELD REVISION NOTES & CORE SLOs

Chapter 1: Introduction to Biology:

Major divisions (Botany, Zoology, Microbiology), Branches (Morphology, Anatomy, Histology, Physiology, Genetics, Biotechnology, Ecology), Relationship with other sciences, Careers, Holy Quran and biological concepts, Muslim scientists (Jabir Bin Hayan, Abdul Malik Asmai, Bu Ali Sina), Levels of Biological Organization (Subatomic to Biosphere).

Chapter 2: Solving a Biological Problem:

Biological Method steps: Observation (Quantitative vs Qualitative), Hypothesis formulation, Deductions (If... then logic), Experimentation, Control vs Experimental Group, Summarization of results, Theory and Law.

Chapter 3: Biodiversity:

Definition & importance of biodiversity, Aims of Classification, History of classification (Aristotle, Abu Usman Umer Al-Jahiz), Two-Kingdom to Five-Kingdom System (Robert Whittaker, Margulis & Schwartz), Binomial Nomenclature (Carl Linnaeus rules), Conservation of biodiversity, Endangered species of Pakistan (Indus Dolphin, Markhor, Houbara Bustard).

Chapter 4: Cells and Tissues:

Microscopy (Magnification vs Resolution, Light vs Electron microscope), Cell Theory (Schleiden, Schwann, Virchow), Prokaryotic vs Eukaryotic cells, Organelles: Nucleus, Mitochondria, Ribosomes, Endoplasmic Reticulum, Golgi apparatus, Plastids, Vacuoles, Cell wall, Cell membrane (Fluid Mosaic Model), Tissues: Plant (Meristematic, Permanent) and Animal (Epithelial, Connective, Muscle, Nervous).

Chapter 5: Cell Cycle:

Interphase (G1, S, G2 phases), Mitosis: Karyokinesis (Prophase, Metaphase, Anaphase, Telophase) and Cytokinesis, Significance of mitosis, Meiosis: Meiosis I (Prophase I crossing over, chiasmata), Significance of meiosis, Necrosis vs Apoptosis.

Chapter 6: Enzymes:

Definition, Active Site, Substrate, Co-factors and Co-enzymes, Characteristics of enzymes, Factors affecting rate: Temperature (Optimum temp, Denaturation), pH, Substrate concentration, Models: Lock and Key Model (Emil Fischer) and Induced Fit Model (Daniel Koshland).

Chapter 7: Bioenergetics:

Oxidation-Reduction in living organisms, ATP as Energy Currency (adenosine + 3 phosphate bonds), Photosynthesis: Light Reaction (Z-scheme, Photolysis, ATP & NADPH formation) and Dark Reaction (Calvin Cycle), Limiting factors of photosynthesis, Cellular Respiration: Aerobic vs Anaerobic (Alcoholic and Lactic acid fermentation), Glycolysis, Krebs cycle, Electron Transport Chain.

Chapter 8: Nutrition:

Autotrophic vs Heterotrophic, Mineral nutrition in plants (Nitrogen, Magnesium deficiency), Components of human food (Carbohydrates, Lipids, Proteins, Minerals, Vitamins A, C, D), Malnutrition (PEM: Marasmus, Kwashiorkor), Human Alimentary Canal: Mouth, Pharynx, Esophagus (Peristalsis), Stomach (Pepsin, Gastric juice), Small Intestine (Duodenum, Jejunum, Ileum, Villi absorption), Large Intestine, Liver & Pancreas functions.

Chapter 9: Transport:

Transport in Plants: Water and mineral uptake by roots, Transpiration (factors affecting rate), Transpiration as a necessary evil, Transpiration Pull Theory, Translocation of organic solutes (Pressure Flow Mechanism), Transport in Humans: Blood components (Plasma, RBCs, WBCs, Platelets), ABO & Rh blood group systems, Human Heart structure and cardiac cycle, Blood vessels (Arteries, Veins, Capillaries), Cardiovascular disorders (Atherosclerosis, Arteriosclerosis, Myocardial Infarction).

SECTION 3: MSNS SENIOR FACULTY BOARD EXAMINATION STRATEGY

- * Draw large, clearly labeled anatomical diagrams with neat horizontal callout pointers.
- * Highlight key terms like Peristalsis, Photolysis, Crossing Over, and Lock-and-Key.
- * Distinguish clearly between terms with tabular side-by-side comparisons (e.g. Mitosis vs Meiosis, Arteries vs Veins).